QUESTION ANALYSIS

EEE475

RF and Microwave Engineering

Question Analysis

Sessions 2018–19 to 2020–21 *3 papers, 77 questions, 138 formulas*

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1. At a glance:

This analysis covers **3 papers** (2018-19, 2019-20, 2020-21), with 77 questions and 452 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

A short-notes question is split across its notes, each taking an equal share of the marks, so some marks are not whole numbers.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Antennas	145	32 %	23	Hertzian dipole (25); Linear arrays (16); Radiation from accelerating charges (13)
Transmission lines	86	19 %	13	Terminated lines (30); Line equations (17); Line constants (11)
Impedance matching	43	10 %	7	Single-stub matching (28); Quarter-wave transformer (15)
Microwave systems	42	9 %	8	Radar (13); Radio link budget (11); Receivers and mixers (6)
Smith chart	28	6 %	4	Constant-r circles (18); Circle properties (6); Reading impedance (4)

A formula name in bold is a key formula.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2018-19	2019-20	2020-21	All
EM waves	·	·	**6**	**6**
Plane waves	·	·	**6**	6
Transmission lines	**21**	**34**	**31**	**86**
Terminated lines	**14**	**4**	**12**	30
Line equations	·	**9**	**8**	17
Line constants	·	·	**11**	11
Lossy lines	·	**10**	·	10
Open and short tests	**7**	·	·	7
Slotted-line measurement	·	**7**	·	7
Reflection coefficient	·	**2**	·	2
Standing wave ratio	·	**2**	·	2
Smith chart	**4**	**12**	**12**	**28**
Constant-r circles	·	**6**	**12**	18
Circle properties	·	**6**	·	6
Reading impedance	**4**	·	·	4
Waveguides and microstrip	**6**	**9**	**12**	**27**
Microstrip	**6**	·	**5**	11
Rectangular waveguide modes	·	·	**7**	7
TM waves	·	**6**	·	6
Circular waveguide modes	·	**3**	·	3
Network analysis	·	**11**	·	**11**
S-parameters	·	**6**	·	6
Signal flow graphs	·	**5**	·	5
Impedance matching	**14**	**10**	**19**	**43**
Single-stub matching	**8**	**6**	**14**	28
Quarter-wave transformer	**6**	**4**	**5**	15

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Topic and subtopic	2018-19	2019-20	2020-21	All
Resonators	·	**2**	**3**	**5**
Cavity resonators	·	**2**	**3**	5
RF components and filters	**12**	**12**	·	**24**
RF behaviour of L and C	**6**	**6**	·	12
Two-port parameters	**6**	·	·	6
Filters	·	**6**	·	6
Amplifier design	**12**	**14**	·	**26**
Constant-gain circles	·	**9**	·	9
Oscillators	**7**	·	·	7
Stability	**5**	·	·	5
Constant-gain condition	·	**5**	·	5
Microwave tubes and devices	·	·	**9**	**9**
Klystron	·	·	**2.25**	2.25
Magnetron	·	·	**2.25**	2.25
Gunn diode	·	·	**2.25**	2.25
Travelling wave tube	·	·	**2.25**	2.25
Microwave systems	**17**	**9**	**16**	**42**
Radar	**6**	**7**	·	13
Radio link budget	**11**	·	·	11
Receivers and mixers	·	·	**6**	6
AM power	·	·	**5**	5
EIRP	·	·	**5**	5
RF hazards	·	**2**	·	2
Antennas	**64**	**39**	**42**	**145**
Hertzian dipole	·	·	**25**	25
Linear arrays	**9**	**7**	·	16
Radiation from accelerating charges	·	·	**13**	13
Antenna aperture	**6**	**6**	·	12
Near and far field	**5**	**7**	·	12
Pattern calculations	**6**	**5**	·	11
Remote sensing	**5**	**5**	·	10
Directivity and beam area	·	**9**	·	9
Magnetic dipole	**8**	·	·	8
Horn antennas	**7**	·	·	7
Paraboloid reflectors	**6**	·	·	6
Effective height and area	·	·	**4**	4
Reciprocity	**3**	·	·	3
Image theory	**3**	·	·	3
Polarization	**3**	·	·	3
Yagi-Uda antenna	**3**	·	·	3

3. Priority list:

Must know: Antennas (145), Transmission lines (86), Impedance matching (43), Microwave systems (42), Smith chart (28), Waveguides and microstrip (27)

Likely: Amplifier design (26), RF components and filters (24), Resonators (5)

Less often: Network analysis (11), Microwave tubes and devices (9), EM waves (6)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. EM waves:

6 marks, 1 question, in 1 of 3 papers. Book: Pozar Ch. 1; Sadiku Ch. 10.

Most often numerical (1 of 1). Subtopics: Plane waves.

Formulas:

Subtopic	Formula	Equation	Used in
Plane waves	**Wave speed**	$u = \dfrac{c}{\sqrt{\mu_r \epsilon_r}}$	20-21 3(a)
	Intrinsic impedance	$\eta = \eta_0 \sqrt{\dfrac{\mu_r}{\epsilon_r}}$	20-21 3(a)
	Wavelength	$\lambda = \dfrac{2\pi}{\beta}$	20-21 3(a)

Questions:

No.	Paper	Marks	Type	Question
Plane waves				
1	20-21 3(a)	6	numerical	A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find: (i) The direction of wave propagation (ii) λ, f, and ϵ_r (iii) $\vec{H}$

4.2. Transmission lines:

86 marks, 13 questions, in 3 of 3 papers. Book: Pozar Ch. 2; Sadiku Ch. 11.

Most often numerical (6 of 13). Subtopics: Terminated lines; Open and short tests; Lossy lines; Reflection coefficient; Standing wave ratio; Line equations; Slotted-line measurement; Line constants.

Formulas:

Subtopic	Formula	Equation	Used in
Terminated lines	**Reflection coefficient**	$\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$	18-19 3(c), 19-20 6(b), 20-21 1(c)
	Standing wave ratio	$S = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	18-19 3(c), 19-20 4(a), 20-21 1(c)
	Voltage maximum	$d_{max} = \dfrac{\theta\lambda}{4\pi}$	18-19 3(c)
	Input impedance	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	18-19 3(c)
	Short-circuited line	$V = -2jV_0^+ \sin\beta z,\ I = \dfrac{2V_0^+}{Z_0}\cos\beta z$	18-19 4(c)
	Short and open input	$Z_{sc} = jZ_0 \tan\beta l,\ Z_{oc} = -jZ_0 \cot\beta l$	18-19 4(c), 18-19 5(d), 19-20 6(c)

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Subtopic	Formula	Equation	Used in
	Load from Γ	$Z_L = Z_0 \dfrac{1+\Gamma}{1-\Gamma}$	20-21 1(c)
	Input impedance	$Z_{in}(l) = \left(\dfrac{Z_L - jZ_0 \tan\beta l}{Z_0 - jZ_L \tan\beta l}\right) Z_0$	20-21 1(c)
Open and short tests	Z_0 from two measurements	$Z_0 = \sqrt{Z_{oc} Z_{sc}}$	18-19 5(d), 19-20 6(c)
	βl from two measurements	$\tan^2 \beta l = -Z_{sc}/Z_{oc}$	18-19 5(d)
Lossy lines	Lossy input impedance	$Z_{in} = Z_0 \dfrac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l}$	19-20 2(d)
	γ from a short	$\gamma l = \dfrac{1}{2} \ln \dfrac{1+T}{1-T},\ T = \tanh\gamma l = \dfrac{Z_{sc}}{Z_0}$	19-20 2(d)
Reflection coefficient	Along the line	$\Gamma(z) = \Gamma_L e^{2\gamma z},\ \Gamma(d) = \Gamma_L e^{-j2\beta d}$	19-20 4(a)
	Return loss	$RL = -20 \log_{10}\lvert\Gamma\rvert$ dB	19-20 4(a)
Line equations	Telegrapher's equations	$\dfrac{dV}{dz} = -(R + j\omega L)\, I,\ \dfrac{dI}{dz} = -(G + j\omega C)\, V$	19-20 6(b), 20-21 1(a)
	General solution	$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$	19-20 6(b), 20-21 1(a)
	Line current	$I(z) = \dfrac{1}{Z_0}\left(V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}\right)$	19-20 6(b)
	Propagation constant	$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$	19-20 6(b), 20-21 1(a)
	Characteristic impedance	$Z_0 = \dfrac{R + j\omega L}{\gamma} = \sqrt{\dfrac{R + j\omega L}{G + j\omega C}}$	19-20 6(b), 20-21 3(b)
	Low-loss line	$\alpha \approx \dfrac{R}{2Z_0} + \dfrac{GZ_0}{2},\ \beta \approx \omega\sqrt{LC},\ Z_0 \approx \sqrt{\dfrac{L}{C}}$	20-21 1(a)
Slotted-line measurement	Phase from the minimum	$\theta_\Gamma = 2\beta d_{min} - \pi$	19-20 6(d)
	Impedance at extrema	$Z_{min} = \dfrac{Z_0}{S},\ Z_{max} = Z_0 S$	19-20 6(d)
	Load from SWR	$Z_L = Z_0 \dfrac{1 - jS \tan\beta d_{min}}{S - j \tan\beta d_{min}}$	19-20 6(d)
Line constants	Conductor loss	$\alpha_c = \dfrac{R}{2Z_0},\ \alpha_d = \alpha - \alpha_c$	20-21 1(b)
	Shunt conductance	$G = \dfrac{2(\alpha - \alpha_c)}{Z_0}$	20-21 1(b)
	Phase velocity	$u_p = \dfrac{\omega}{\beta}$	20-21 3(b)

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Questions:

No.	Paper	Marks	Type	Question
Terminated lines				
1	18-19 3(c)	7	numerical	A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line.
2	18-19 4(c)	7	derivation	Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length.
3	19-20 6(c)	4	theory	Briefly describe input impedance for open circuit termination and short circuit termination.
4	20-21 1(c)	12	derivation	In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan \beta l}{Z_0 - jZ_L \tan \beta l}\right) Z_0$
Open and short tests				
5	18-19 5(d)	7	numerical	A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6$ Ω and short circuit impedance $j103$ Ω. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input.
Lossy lines				
6	19-20 2(d)	10	numerical	The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit *[sic: should be short-circuit]* is replaced by a load impedance $Z_L = 37 + j45$ Ω.
Reflection coefficient				
7	19-20 4(a)i	2	theory	Write short notes on the following topic (i) Reflection coefficient
Standing wave ratio				
8	19-20 4(a)ii	2	theory	Write short notes on the following topic (ii) Standing wave ratio
Line equations				
9	19-20 6(b)	9	derivation	From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient.
10	20-21 1(a)	8	derivation	From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line.
Slotted-line measurement				
11	19-20 6(d)	7	numerical	The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L.
Line constants				
12	20-21 1(b)	5	numerical	If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4$ Ω m^{-1}, $L = 0.5$ μH m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit.
13	20-21 3(b)	6	numerical	A telephone line has $R = 30$ Ω km^{-1}, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20$ μF km^{-1}. At $f = 1$ kHz, obtain: (i) The characteristic impedance of the line (ii) The propagation constant (iii) The phase velocity

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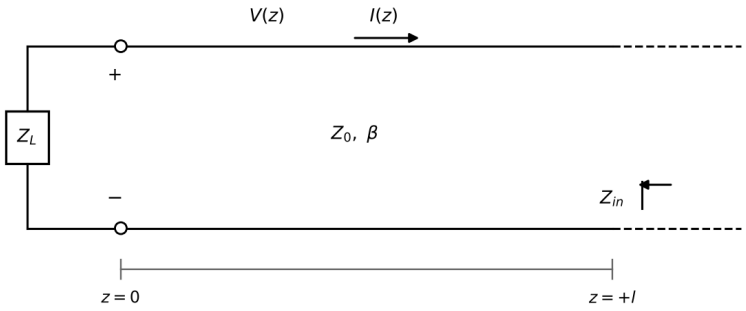


Figure 1: 20-21 1(c): Fig. 1 of the paper (redrawn)

4.3. Smith chart:

28 marks, 4 questions, in 3 of 3 papers. Book: Ludwig and Bretchko Ch. 3.

Most often theory (2 of 4). Subtopics: Reading impedance; Constant-r circles; Circle properties.

Formulas:

Subtopic	Formula	Equation	Used in
Reading impedance	**Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$	18-19 3(b)
Constant-r circles	Impedance and Γ	$z = r + jx = \dfrac{1+\Gamma}{1-\Gamma}$	19-20 2(a)
	Constant-r circle	$\left(u - \dfrac{r}{1+r}\right)^2 + v^2 = \left(\dfrac{1}{1+r}\right)^2$	19-20 2(a), 20-21 2(a)
	Normalized impedance	$z = \dfrac{Z}{Z_0} = r + jx,\ \Gamma = \dfrac{z-1}{z+1}$	20-21 2(a)
Circle properties	**Constant-x circle**	$(u-1)^2 + \left(v - \dfrac{1}{x}\right)^2 = \left(\dfrac{1}{x}\right)^2$	19-20 2(b), 20-21 2(a)

Questions:

No.	Paper	Marks	Type	Question
Reading impedance				
1	18-19 3(b)	4	theory	Write the steps to determine line input impedance using Smith Chart.
Constant-r circles				
2	19-20 2(a)	6	derivation	Show that the normalized resistances represent circles in reflection coefficient plane.
3	20-21 2(a)	12	derivation repeats 19-20 2(a)	Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load.
Circle properties				
4	19-20 2(b)	6	theory	Write the properties of r-circles and x-circles of a Smith chart.

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4.4. Waveguides and microstrip:

27 marks, 5 questions, in 3 of 3 papers. Book: Pozar Ch. 3; Sadiku Ch. 12.

Most often theory (3 of 5). Subtopics: Microstrip; Circular waveguide modes; TM waves; Rectangular waveguide modes.

Formulas:

Subtopic	Formula	Equation	Used in
Circular waveguide modes	TE wave impedance	$Z_{TE} = \dfrac{k\eta}{\beta}$	19-20 2(c)
	Circular guide cutoff	$f_c = \dfrac{p\,c}{2\pi a\sqrt{\epsilon_r}}$	19-20 2(c)
TM waves	TM field	$E_z = E_0 \sin\dfrac{m\pi x}{a} \sin\dfrac{n\pi y}{b}\, e^{-j\beta z}$	19-20 5(a)
	Cutoff wave number	$k_c = \sqrt{\left(\dfrac{m\pi}{a}\right)^2 + \left(\dfrac{n\pi}{b}\right)^2},\ k_c^2 = k^2 - \beta^2$	19-20 5(a)
	Phase constant	$\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)
	Cutoff frequency	$f_{c,mn} = \dfrac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$	19-20 5(a), 20-21 3(d)
	TM wave impedance	$Z_{TM} = \dfrac{E_x}{H_y} = \dfrac{\beta}{\omega\epsilon} = \eta\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)

Questions:

No.	Paper	Marks	Type	Question
Microstrip				
1	18-19 4(b)	6	theory repeats 20-21 6(a)	What is microstrip antenna? Mention some of the applications of microstrip line.
2	20-21 6(a)	5	theory	What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line.
Circular waveguide modes				
3	19-20 2(c)	3	theory	Describe how TE mode is different from TM mode in a circular waveguide.
TM waves				
4	19-20 5(a)	6	derivation	Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave.
Rectangular waveguide modes				
5	20-21 3(d)	7	numerical	A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes.

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4.5. Network analysis:

11 marks, 2 questions, in 1 of 3 papers. Book: Pozar Ch. 4.

Most often theory (1 of 2). Subtopics: S-parameters; Signal flow graphs.

Formulas:

Subtopic	Formula	Equation	Used in
S-parameters	Power waves	$a_n = \frac{V_n^+}{\sqrt{Z_0}},\ b_n = \frac{V_n^-}{\sqrt{Z_0}}$	19-20 3(a)
	S-parameters	$[b] = [S][a],\ S_{ij} = \left.\frac{b_i}{a_j}\right\|_{a_k=0,\, k\neq j}$	19-20 3(a)
	Ideal circulator	$S_{21} = S_{32} = S_{13} = 1$, all other $S_{ij} = 0$	19-20 3(a)
Signal flow graphs	**Input impedance**	$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	19-20 3(c)
	Input reflection	$\Gamma_{in} = \Gamma_L e^{-j2\beta l}$	19-20 3(c)
	Flow graph with the source	$a_1 = \frac{b_s}{1 - \Gamma_S \Gamma_L e^{-j2\beta l}}$	19-20 3(c)

Questions:

No.	Paper	Marks	Type	Question
S-parameters				
1	19-20 3(a)	6	theory	Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example.
Signal flow graphs				
2	19-20 3(c)	5	derivation	For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation.

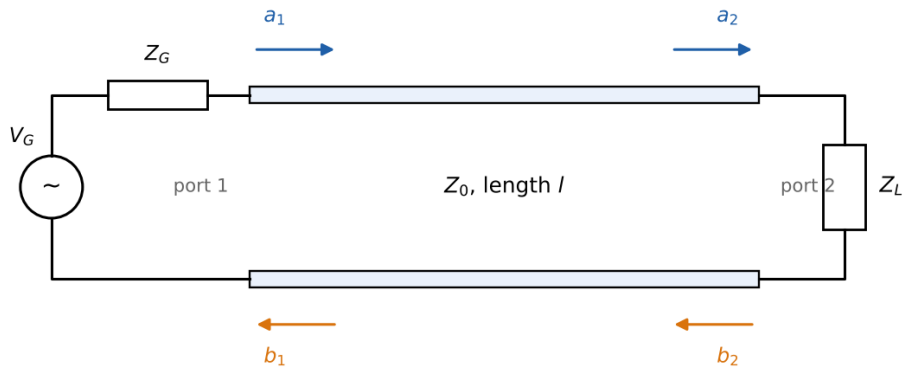


Figure 2: 19-20 3(c): Figure 1 of the paper (redrawn)

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4.6. Impedance matching:

43 marks, 7 questions, in 3 of 3 papers. Book: Pozar Ch. 5.

Most often design (4 of 7). Subtopics: Single-stub matching; Quarter-wave transformer.

Formulas:

Subtopic	Formula	Equation	Used in
Single-stub matching	Stub position	$t = \tan\beta d = \dfrac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]/Z_0}}{R_L - Z_0}$	18-19 3(d), 19-20 4(d)
	Short-circuited stub	$\dfrac{l}{\lambda} = \dfrac{1}{2\pi}\tan^{-1}\dfrac{Y_0}{B}$	18-19 3(d), 19-20 4(d), 20-21 2(b), 20-21 3(c)
	Susceptance to cancel	$B = \dfrac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0\,[R_L^2 + (X_L + Z_0 t)^2]}$	19-20 4(d)
Quarter-wave transformer	**Quarter-wave transformer**	$Z_1 = \sqrt{Z_0 R_L}$	18-19 4(d), 19-20 4(b), 20-21 2(c)

Questions:

No.	Paper	Marks	Type	Question
Single-stub matching				
1	18-19 3(d)	8	design repeats 19-20 4(d)	A 50 Ω transmission line is connected to a load impedance $35 - j47.5$ Ω. Find the position and length of a short circuited stub required to match the line.
2	19-20 4(d)	6	design	A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line.
3	20-21 2(b)	8	design repeats 19-20 4(d)	A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)$ Ω. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart.
4	20-21 3(c)	6	design	An antenna with an impednace of $40 + j30$ Ω is to be matched to a 100 Ω lossless line with a shorted stub. Determine: (i) The required stub admittance (ii) The distance between the stub and the antenna (iii) The stub length (iv) The standing wave ratio on each segment of the system
Quarter-wave transformer				
5	18-19 4(d)	6	theory repeats 19-20 4(b)	How is quarter wavelength transmission line used for impedance matching?
6	19-20 4(b)	4	theory	Show how quarter wavelength line is used for matching.
7	20-21 2(c)	5	theory repeats 19-20 4(b)	Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations.

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4.7. Resonators:

5 marks, 2 questions, in 2 of 3 papers. Book: Pozar Ch. 6.

Most often theory (1 of 2). Subtopics: Cavity resonators.

Formulas:

Subtopic	Formula	Equation	Used in
Cavity resonators	**Cavity resonance**	$f_{mnp} = \frac{c}{2\sqrt{\mu_r \epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$	19-20 4(a), 20-21 6(c)
	Quality factor	$Q = \omega \frac{\text{energy stored}}{\text{power lost}}$	19-20 4(a)

Questions:

No.	Paper	Marks	Type	Question
Cavity resonators				
1	19-20 4(a)iv	2	theory	Write short notes on the following topic (iv) Microwave cavity resonators
2	20-21 6(c)	3	numerical	A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$.

4.8. RF components and filters:

24 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 1, 4, 5.

Most often derivation (2 of 4). Subtopics: RF behaviour of L and C; Two-port parameters; Filters.

Formulas:

Subtopic	Formula	Equation	Used in
RF behaviour of L and C	High-frequency resistor	$Z = j\omega(2L) + \frac{1}{1/R + j\omega C_a}$	18-19 4(a)
	Resistor corners	$f_c = \frac{1}{2\pi R C_a},\ f_0 = \frac{1}{2\pi\sqrt{2LC_a}}$	18-19 4(a)
	Ideal reactances	$X_L = 2\pi f L,\ X_C = \frac{1}{2\pi f C}$	19-20 6(a)
	Capacitor self-resonance	$f_r = \frac{1}{2\pi\sqrt{L_s C}}$	19-20 6(a)
	Inductor self-resonance	$f_r = \frac{1}{2\pi\sqrt{L C_s}}$	19-20 6(a)
Two-port parameters	ABCD definition	$V_1 = AV_2 + BI_2,\ I_1 = CV_2 + DI_2$	18-19 5(a)
	High-pass ABCD	$A = 1 - \frac{1}{\omega^2 LC},\ B = \frac{1}{j\omega C},\ C = \frac{1}{j\omega L},\ D = 1$	18-19 5(a)
Filters	**High-pass gain**	$H = \frac{V_2}{V_1} = \frac{j\omega RC}{1 + j\omega RC},\ f_c = \frac{1}{2\pi RC}$	19-20 5(b)
	Gain and phase	$\lvert H\rvert = \frac{f/f_c}{\sqrt{1 + (f/f_c)^2}},\ \angle H = 90° - \tan^{-1}\frac{f}{f_c}$	19-20 5(b)
	Series C in a Z_0 system	$S_{21} = \frac{j\omega(2Z_0C)}{1 + j\omega(2Z_0C)}$	19-20 5(b)

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Questions:

No.	Paper	Marks	Type	Question
RF behaviour of L and C				
1	18-19 4(a)	6	diagram	Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure *[sic: should be curve]* of the resistor.
2	19-20 6(a)	6	theory	Explain how radio frequency affects the inductor and capacitor reactance.
Two-port parameters				
3	18-19 5(a)	6	derivation	For high pass filter determine ABCD parameters of a two port system.
Filters				
4	19-20 5(b)	6	derivation	Find the forward voltage gain of a high-pass filter.

4.9. Amplifier design:

26 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 9, 10; Pozar Ch. 12, 13.

Most often theory (3 of 4). Subtopics: Stability; Oscillators; Constant-gain condition; Constant-gain circles.

Formulas:

Subtopic	Formula	Equation	Used in
Stability	Input reflection	$\Gamma_{in} = S_{11} + \dfrac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$	18-19 5(b)
	Output stability circle	$C_L = \dfrac{(S_{22} - \Delta S_{11}^*)^*}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2},\ R_L = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
	Input stability circle	$C_S = \dfrac{(S_{11} - \Delta S_{22}^*)^*}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2},\ R_S = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
Oscillators	Negative-resistance oscillator	$R_{in} + R_L = 0,\ X_{in} + X_L = 0$	18-19 5(c)
	Barkhausen condition	$\lvert A\beta\rvert = 1,\ \angle A\beta = 0°$	18-19 5(c)
Constant-gain condition	Unilateral gain	$G_{TU} = G_S\, G_0\, G_L,\ G_0 = \lvert S_{21}\rvert^2$	19-20 3(b)
	Source gain	$G_S = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$	19-20 3(b)
	Normalized gain	$g_s = \dfrac{G_S}{G_{Smax}} = G_S(1 - \lvert S_{11}\rvert^2)$	19-20 3(b)
	Gain-circle centre	$d_{gs} = \dfrac{g_s S_{11}^*}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
	Gain-circle radius	$r_{gs} = \dfrac{\sqrt{1 - g_s}\,(1 - \lvert S_{11}\rvert^2)}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
Constant-gain circles	**Maximum source gain**	$G_{Smax} = \dfrac{1}{1 - \lvert S_{11}\rvert^2}$	19-20 3(d)

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Questions:

No.	Paper	Marks	Type	Question
Stability				
1	18-19 5(b)	5	theory	What are stability circles? Explain how it is used.
Oscillators				
2	18-19 5(c)	7	theory	Explain the working principle of negative resistance oscillator and feedback oscillator.
Constant-gain condition				
3	19-20 3(b)	5	theory	Explain the condition for constant gain for amplifier.
Constant-gain circles				
4	19-20 3(d)	9	design	A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S.

4.10. Microwave tubes and devices:

9 marks, 4 questions, in 1 of 3 papers. Book: Pozar Ch. 11.

Most often theory (4 of 4). Subtopics: Klystron; Magnetron; Gunn diode; Travelling wave tube.

Questions:

No.	Paper	Marks	Type	Question
Klystron				
1	20-21 5(d)i	2.25	theory	Write short notes (any three). (i) Klystron
Magnetron				
2	20-21 5(d)ii	2.25	theory	Write short notes (any three). (ii) Magnetron
Gunn diode				
3	20-21 5(d)iii	2.25	theory	Write short notes (any three). (iii) Gunn Diode Oscillator
Travelling wave tube				
4	20-21 5(d)iv	2.25	theory	Write short notes (any three). (iv) Travelling Wave Tube (TWT)

4.11. Microwave systems:

42 marks, 8 questions, in 3 of 3 papers. Book: Pozar Ch. 14.

Most often theory (3 of 8). Subtopics: Radar; Radio link budget; RF hazards; Receivers and mixers; AM power; EIRP.

Formulas:

Subtopic	Formula	Equation	Used in
Radar	**Radar range**	$R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$	18-19 2(b), 19-20 4(c)
	Density at the target	$S_t = \frac{P_t G}{4\pi R^2}$	19-20 4(c)
	Radar equation	$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$	19-20 4(c)
	Range from delay	$R = \frac{c\,t}{2}$	19-20 4(c)
Radio link budget	**Directivity from aperture**	$D = \frac{4\pi A_e}{\lambda^2}$	18-19 6(c)

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Subtopic	Formula	Equation	Used in
	Friis formula	$\frac{P_r}{P_t} = \frac{A_{et}A_{er}}{r^2\lambda^2} = G_tG_r\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c), 18-19 6(c)
	Noise power	$N = kT_{sys}B$	18-19 2(c)
	Link SNR	$\frac{S}{N} = \frac{P_tG_tG_r}{kT_{sys}B}\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c)
RF hazards	Specific absorption rate	$\text{SAR} = \frac{\sigma_t\lvert E\rvert^2}{\rho}$	19-20 4(a)
	Safe distance	$r = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}}$	19-20 4(a)
Receivers and mixers	Mixer product	$\cos\omega_{RF}t\cos\omega_{LO}t = \frac{1}{2}\cos(\omega_{RF}-\omega_{LO})t + \frac{1}{2}\cos(\omega_{RF}+\omega_{LO})t$	20-21 5(a)
	Intermediate frequency	$f_{IF} = f_{RF} - f_{LO}$	20-21 5(a)
	Image frequency	$f_{image} = f_{LO} - f_{IF}$	20-21 5(a)
AM power	AM signal	$s(t) = A_c[1+\mu\cos\omega_m t]\cos\omega_c t$	20-21 5(b)
	AM total power	$P_t = P_c\left(1+\frac{\mu^2}{2}\right)$	20-21 5(b)
EIRP	Gain in dB	$G_{dB} = 10\log_{10}G,\ G = 10^{G_{dB}/10}$	20-21 5(c)
	EIRP	$\text{EIRP} = P_tG_t$	20-21 5(c)

Questions:

No.	Paper	Marks	Type	Question
Radar				
1	18-19 2(b)	6	theory repeats 19-20 4(c)	Explain how the RADAR range equation is used to detect objects.
2	19-20 4(c)	7	theory	Explain how the RADAR range equation is used to detect object.
Radio link budget				
3	18-19 2(c)	5	derivation	Show the signal to noise ratio of a radio link is related to bandwidth and system temperature.
4	18-19 6(c)	6	derivation	Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them.
RF hazards				
5	19-20 4(a)iii	2	theory	Write short notes on the following topic (iii) RF radiation hazards and safety standard
Receivers and mixers				
6	20-21 5(a)	6	numerical	An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz. (i) Calculate the IF (intermediate frequency) output. (ii) Why are active filters used after mixing in a super-heterodyne receiver?
AM power				
7	20-21 5(b)	5	numerical	For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W.
EIRP				
8	20-21 5(c)	5	numerical	For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna.

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4.12. Antennas:

145 marks, 23 questions, in 3 of 3 papers. Book: Kraus; Sadiku Ch. 13.

Most often theory (12 of 23). Subtopics: Antenna aperture; Pattern calculations; Near and far field; Magnetic dipole; Remote sensing; Linear arrays; Reciprocity; Image theory; Polarization; Yagi-Uda antenna; Paraboloid reflectors; Horn antennas; Directivity and beam area; Hertzian dipole; Radiation from accelerating charges; Effective height and area.

Formulas:

Subtopic	Formula	Equation	Used in
Antenna aperture	**Directivity from aperture**	$D = \dfrac{4\pi A_e}{\lambda^2}$	18-19 1(a), 18-19 6(d)
	Effective aperture	$A_e = \dfrac{P_r}{S},\ \varepsilon_{ap} = \dfrac{A_e}{A_p}$	19-20 1(b)
	Aperture and directivity	$D = \dfrac{4\pi A_e}{\lambda^2},\ A_e\,\Omega_A = \lambda^2$	19-20 1(b)
	Aperture beamwidth	$\text{HPBW} \approx \dfrac{51^\circ\,\lambda}{a}$	19-20 1(b)
Pattern calculations	Power and beam area	$P_{rad} = \iint U\,d\Omega,\ \Omega_A = \dfrac{P_{rad}}{U_m}$	18-19 1(b)
	Directivity	$D = \dfrac{4\pi U_m}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$	18-19 1(b), 19-20 1(a)
	Radiated power	$P_{rad} = \displaystyle\int_0^{2\pi}\int_0^{\pi} U\,\sin\theta\,d\theta\,d\phi$	19-20 1(d)
Near and far field	Field regions	$r_1 = 0.62\sqrt{D^3/\lambda},\ r_2 = 2D^2/\lambda$	18-19 1(c), 19-20 5(d)
Magnetic dipole	Magnetic dipole E	$E_\phi = \dfrac{\eta k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_r	$H_r = \dfrac{jkIS}{2\pi r^2}\cos\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_θ	$H_\theta = -\dfrac{k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr} - \dfrac{1}{(kr)^2}\right]e^{-jkr}$	18-19 1(d)
Remote sensing	Brightness temperature	$T_B = e\,T_{phys}$	18-19 2(a)
	Received noise power	$P = kT_AB$	18-19 2(a), 19-20 1(c)
	Brightness temperature	$T_B = e\,T_{phys},\ e = 1 - \lvert\Gamma\rvert^2$	19-20 1(c)
	Antenna temperature	$T_A = \dfrac{1}{\Omega_A}\iint T_B\,P_n\,d\Omega$	19-20 1(c)
	Radiometer resolution	$\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$	19-20 1(c)
Linear arrays	Phase difference	$\psi = \beta d\cos\theta + \delta$	18-19 2(d), 19-20 5(c)
	Array factor	$\lvert AF_n\rvert = \left\lvert\dfrac{\sin(N\psi/2)}{N\sin(\psi/2)}\right\rvert$	18-19 2(d), 19-20 5(c)

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Subtopic	Formula	Equation	Used in
	Main beam	$\cos\theta_m = -\dfrac{\delta}{\beta d}$	18-19 2(d), 19-20 5(c)
	Nulls	$\theta_k = \cos^{-1}\left[\dfrac{1}{\beta d}\left(-\delta \pm \dfrac{2k\pi}{N}\right)\right]$	18-19 2(d), 19-20 5(c)
	Broadside beamwidths	$\text{FNBW} \approx \dfrac{2\lambda}{Nd},\ \text{HPBW} \approx 0.886\dfrac{\lambda}{Nd}$	18-19 2(d)
	Broadside beamwidths	$\text{FNBW} = 2\left[\dfrac{\pi}{2} - \cos^{-1}\dfrac{\lambda}{Nd}\right],\ \text{HPBW} \approx 0.886\,\dfrac{\lambda}{Nd}$	19-20 5(c)
	End-fire beamwidth	$\text{FNBW} = 2\cos^{-1}\left(1 - \dfrac{\lambda}{Nd}\right)$	19-20 5(c)
Polarization	Polarization match	$F = \cos^2\gamma$	18-19 6(a)
Paraboloid reflectors	Paraboloid shape	$r = \dfrac{2f}{1+\cos\theta}$	18-19 6(b)
	Paraboloid taper	$\dfrac{S(\theta)}{S(0)} = \left(\dfrac{1+\cos\theta}{2}\right)^2,\ \dfrac{E(\theta)}{E(0)} = \dfrac{1+\cos\theta}{2}$	18-19 6(b)
Horn antennas	**Horn directivity**	$D \approx \dfrac{7.5A_p}{\lambda^2}$	18-19 6(d)
Directivity and beam area	Directivity estimate	$D \approx \dfrac{41\,253}{\theta_{HP}\,\phi_{HP}}$	19-20 1(a)
	Gain	$G = k\,D$	19-20 1(a)
	Beam area	$\Omega_A = \iint P_n(\theta,\phi)\sin\theta\,d\theta\,d\phi \approx \theta_{HP}\,\phi_{HP}$	19-20 1(a)
	Effective height	$h_e = \dfrac{V_{oc}}{E} = \dfrac{1}{I_0}\displaystyle\int I(z)\,dz$	19-20 1(a), 20-21 6(d)
	Height and aperture	$A_e = \dfrac{h_e^2\,Z_0}{4R_r}$	19-20 1(a)
Hertzian dipole	Vector potential	$A_r = \dfrac{\mu_0 I dl}{4\pi r}e^{-j\beta r}\cos\theta,\ A_\theta = -\dfrac{\mu_0 I dl}{4\pi r}e^{-j\beta r}\sin\theta$	20-21 4(a)
	Magnetic field	$\mathbf{H} = \dfrac{Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2}\right]e^{-j\beta r}\,\hat{a}_\phi$	20-21 4(a)
	Field from **H**	$\mathbf{E} = \dfrac{1}{j\omega\epsilon}\nabla\times\mathbf{H},\ \dfrac{1}{j\omega\epsilon} = \dfrac{\eta}{j\beta}$	20-21 4(b)
	Radial E field	$E_r = \dfrac{\eta Idl}{2\pi}\cos\theta\left[\dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Polar E field	$E_\theta = \dfrac{\eta Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Average Poynting vector	$\mathbf{P}_{avg} = \dfrac{1}{2}\text{Re}\left(\mathbf{E}\times\mathbf{H}^*\right)$	20-21 4(c)
	Far-field power density	$\mathbf{P}_{avg} = \dfrac{15\pi I_0^2}{r^2}\left(\dfrac{dl}{\lambda}\right)^2\sin^2\theta\,\hat{a}_r$	20-21 4(c)

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Subtopic	Formula	Equation	Used in
	Radiated power	$P_{rad} = 40\pi^2 \left(\frac{dl}{\lambda}\right)^2 I_0^2$	20-21 4(c)
	Radiation resistance	$R_{rad} = \frac{2P_{rad}}{I_0^2} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$	20-21 4(c)
Radiation from accelerating charges	Retarded time	$t_r = t - \frac{R}{c}$	20-21 6(b)
	Retarded potentials	$V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(t_r)}{R}\,d\tau',\ \mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(t_r)}{R}\,d\tau'$	20-21 6(b)
	Jefimenko, E	$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2 R}\right] d\tau'$	20-21 6(b)
	Jefimenko, B	$\mathbf{B} = \frac{\mu_0}{4\pi}\int \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR}\right] \times \hat{\mathbf{R}}\,d\tau'$	20-21 6(b)
Effective height and area	**Effective area**	$A_e = \frac{P_{received}}{P_{inc}} = \frac{\lambda^2}{4\pi}G$	20-21 6(d)
	Area from height	$A_e = \frac{\eta\, h_e^2}{4R_{rad}}$	20-21 6(d)

Questions:

No.	Paper	Marks	Type	Question
Antenna aperture				
1	18-19 1(a)	6	theory repeats 19-20 1(b)	What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna.
2	19-20 1(b)	6	theory	What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna.
Pattern calculations				
3	18-19 1(b)	6	numerical repeats 19-20 1(d)	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \le \theta \le \pi$, $0 \le \varphi \le \pi$.
4	19-20 1(d)	5	numerical	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \le \theta \le \pi,\ 0 \le \varphi \le \pi$.
Near and far field				
5	18-19 1(c)	5	theory repeats 19-20 5(d)	Describe the field regions surrounding an antenna.
6	19-20 5(d)	7	theory	Define near field and far field for an antenna.
Magnetic dipole				
7	18-19 1(d)	8	derivation	For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity.
Remote sensing				
8	18-19 2(a)	5	theory repeats 19-20 1(c)	Describe how passive remote sensing is used to detect object.

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No.	Paper	Marks	Type	Question
9	19-20 1(c)	5	theory	Describe how passive remote sensing is used to detect object.
Linear arrays				
10	18-19 2(d)	9	derivation repeats 19-20 5(c)	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
11	19-20 5(c)	7	derivation	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
Reciprocity				
12	18-19 3(a)i	3	theory	Write short notes on. (i) Lorentz reciprocity theorem.
Image theory				
13	18-19 3(a)ii	3	theory	Write short notes on. (ii) Image theory.
Polarization				
14	18-19 6(a)i	3	theory	Write short notes on the following topics. (i) Poincare Sphere
Yagi-Uda antenna				
15	18-19 6(a)ii	3	theory	Write short notes on the following topics. (ii) Yagi-Uda antenna.
Paraboloid reflectors				
16	18-19 6(b)	6	derivation	For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio.
Horn antennas				
17	18-19 6(d)	7	derivation	For a pyramidal or rectangular horn show that directivity D is $D \approx 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right)$.
Directivity and beam area				
18	19-20 1(a)	9	theory	Write short notes on the following topics (i) Directivity (ii) Effective Height (iii) Beam Area
Hertzian dipole				
19	20-21 4(a)	8	derivation	The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r} \hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi} \cos(\theta) \left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r} \hat{\phi}$ *[sic: should be* $\sin(\theta)$*]*.
20	20-21 4(b)	12	derivation	Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta I dl}{4\pi} \cos(\theta) \left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r} \hat{\theta} + \frac{\eta I dl}{2\pi} \cos(\theta) \left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r} \hat{r}$ *[sic: the* $\hat{\theta}$ *term should have* $\sin(\theta)$*]*
21	20-21 4(c)	5	derivation	Find the expression of average power density of Hertzian dipole in far field zone.
Radiation from accelerating charges				
22	20-21 6(b)	13	derivation	Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation.
Effective height and area				
23	20-21 6(d)	4	theory repeats 19-20 1(a)	What do you mean by effective height and effective area of an antenna?

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5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Transmission lines				
1	18-19 4(c)	7	Terminated lines	Apply $V = 0$ at $z = 0$, so $V_0^- = -V_0^+$; use Euler identities for V and I; divide for Z_{in}.
2	19-20 6(b)	9	Line equations	Solve the line equations; take V and I as forward and backward waves; form V/I for Z_0; apply the load for Γ.
3	20-21 1(a)	8	Line equations	Apply KVL and KCL to a short section; let Δz go to zero; combine into the wave equation; read off γ; simplify for low loss.
4	20-21 1(c)	12	Terminated lines	Apply V and I at the load to find Γ; move a distance l along the line; form V/I; rewrite in tan βl.
Smith chart				
5	19-20 2(a)	6	Constant-r circles	Write z = (1+Γ)/(1−Γ) with Γ = u+jv; take the real part; complete the square to get a circle.
6	20-21 2(a)	12	Constant-r circles	Write z = (1+Γ)/(1−Γ) with Γ = u+jv; split real and imaginary parts; complete the square for each circle.
Waveguides and microstrip				
7	19-20 5(a)	6	TM waves	Start from the wave equation for E_z; separate variables; apply the wall conditions; find k_c, β, f_c and Z_{TM}.
Network analysis				
8	19-20 3(c)	5	Signal flow graphs	Draw a node for each wave; add a branch for each path; apply the self-loop rule to find b/a, then Z_{in}.
RF components and filters				
9	18-19 5(a)	6	Two-port parameters	Write the matrix of the series C and of the shunt L; multiply them.
10	19-20 5(b)	6	Filters	Treat the RC filter as a voltage divider; write V_o/V_i; find its size and phase; read off the cut-off.
Microwave systems				
11	18-19 2(c)	5	Radio link budget	Spread the power over a sphere; collect it with the aperture; divide by the noise power $kT_{sys}B$.
12	18-19 6(c)	6	Radio link budget	Density at the receiver with $D = 4\pi A/\lambda^2$; multiply by the receiving aperture.
Antennas				
13	18-19 1(d)	8	Magnetic dipole	Find **A** of the loop by Stokes; curl it for **H**; curl **H** for **E**.
14	18-19 2(d)	9	Linear arrays	Add the fields of N elements; sum the geometric series; set $\psi = 0$ for the main beam and $\psi = \pm 2k\pi/N$ for nulls.
15	18-19 6(b)	6	Paraboloid reflectors	Write r and ρ in terms of θ; balance the power in a cone and a ring; take the ratio to the centre.
16	18-19 6(d)	7	Horn antennas	Put $A_e = 0.6A_p$ in $D = 4\pi A_e/\lambda^2$.
17	19-20 5(c)	7	Linear arrays	Add the fields of N elements; sum the geometric series; set $\psi = 0$ for the main beam and $\psi = 2\pi$k/N for nulls.
18	20-21 4(a)	8	Hertzian dipole	Take the curl of A in spherical coordinates; divide by μ to get H.
19	20-21 4(b)	12	Hertzian dipole	Use the curl of H in free space; divide by j$\omega\varepsilon$; collect the r and θ components of E.
20	20-21 4(c)	5	Hertzian dipole	Keep the 1/r far-field terms; form the Poynting vector E × H*; average it over time.
21	20-21 6(b)	13	Radiation from accelerating charges	Start from the retarded potentials; take E and B from them; keep the 1/R term, which needs an accelerating charge.

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6. Repeat tracker:

Questions that came back, exactly or with new numbers.

Paper	Asked before as	Topic	Subtopic
2018-19 1(a)	2019-20 1(b)	Antennas	Antenna aperture
2018-19 1(b)	2019-20 1(d)	Antennas	Pattern calculations
2018-19 1(c)	2019-20 5(d)	Antennas	Near and far field
2018-19 2(a)	2019-20 1(c)	Antennas	Remote sensing
2018-19 2(b)	2019-20 4(c)	Microwave systems	Radar
2018-19 2(d)	2019-20 5(c)	Antennas	Linear arrays
2018-19 3(d)	2019-20 4(d)	Impedance matching	Single-stub matching
2018-19 4(b)	2020-21 6(a)	Waveguides and microstrip	Microstrip
2018-19 4(d)	2019-20 4(b)	Impedance matching	Quarter-wave transformer
2020-21 2(a)	2019-20 2(a)	Smith chart	Constant-r circles
2020-21 2(b)	2019-20 4(d)	Impedance matching	Single-stub matching
2020-21 2(c)	2019-20 4(b)	Impedance matching	Quarter-wave transformer
2020-21 6(d)	2019-20 1(a)	Antennas	Effective height and area

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
EM waves	·	·	1	·	·
Transmission lines	3	4	6	·	·
Smith chart	2	2	·	·	·
Waveguides and microstrip	3	1	1	·	·
Network analysis	1	1	·	·	·
Impedance matching	3	·	·	·	4
Resonators	1	·	1	·	·
RF components and filters	1	2	·	1	·
Amplifier design	3	·	·	·	1
Microwave tubes and devices	4	·	·	·	·
Microwave systems	3	2	3	·	·
Antennas	12	9	2	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Antennas	145	32 %
2	Transmission lines	86	51 %
3	Impedance matching	43	61 %
4	Microwave systems	42	70 %
5	Smith chart	28	76 %
6	Waveguides and microstrip	27	82 %
7	Amplifier design	26	88 %
8	RF components and filters	24	93 %
9	Network analysis	11	96 %
10	Microwave tubes and devices	9	98 %

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No.	Topic	Marks	Covered so far
11	EM waves	6	99 %
12	Resonators	5	100 %

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